Fraudulent Job Ad Detection

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1. Exploratory Analysis

Dataset Overview

The dataset contains 17,880 job postings scraped from social media and job platforms, with 18 features covering job metadata (title, location, salary), company information (profile, logo), job content (description, requirements, benefits), and a binary `fraudulent` label. The dataset is heavily imbalanced, approximately 14,000 legitimate postings versus 866 fraudulent ones (~5%). A pre-labelled `in_balanced_dataset` column was available to subset a balanced 685-row training set. However, the full dataset was used for training. Undersampling to 685 rows introduces high variance and does not reflect real-world distribution where fraud is rare (~5%), a model trained on a 50:50 split would over-flag legitimate jobs in production.

Data Quality & Limitations

- **Many null values:** department was dropped (~70% null). salary_range had many nulls but was retained as a feature since null salary correlates with fraud.
- **HTML in text columns:** description, company_profile, requirements and benefits all contained raw HTML tags that were stripped using BeautifulSoup before any NLP processing.
- **Location parsing:** location was split into country, region, and city. Only country was retained, region had 79 unique values and city had 190, both too high for one-hot encoding.
- **Class imbalance:** ~95% legitimate vs ~5% fraudulent. Rather than undersampling, `class_weight='balanced'` and `scale_pos_weight` were used to penalise misclassification of the minority class.
- **Inconsistent text quality:** Many job descriptions contain boilerplate or copied text, which may reduce the discriminative power of NLP features.

Key Patterns & Signals

Exploratory analysis of fraudulent vs non-fraudulent job postings revealed several consistent signals:

- **US-based postings (strongest signal):** `country_US` is an important feature as the majority of job postings originate from the US, and the model learns to distinguish fraud patterns within this dominant group.
- **Vague or missing company profile:** Fraudulent postings tend to have absent or poorly written company profiles. NLP zero-shot scoring confirmed low legitimacy scores for these entries.
- **Low education requirements:** Fraudulent ads disproportionately require only 'High School or equivalent' or 'Unspecified', targeting less experienced or more vulnerable job seekers.
- **Missing salary range:** Null salary information is more prevalent in fraudulent postings, legitimate employers typically disclose compensation.
- **Industry concentration:** Fraudulent postings are concentrated in Oil & Energy, Hospital & Health Care, and Marketing.

2. Proposed Approach for Classification

Approach

A supervised binary classification approach was adopted, treating fraud detection as a binary classification problem (fraudulent vs legitimate). Three models were trained and compared: **Random Forest**, **XGBoost**, and **Logistic Regression**. These were chosen for their ability to handle high-dimensional tabular data, their support for class

weighting, and their interpretability via feature importance. The full imbalanced dataset (~14,880 rows after cleaning) was used with an 80:20 stratified train-test split to preserve class proportions in both sets.

Challenges & Mitigations

- **Class imbalance:** Used `class_weight='balanced'` for Random Forest and Logistic Regression, and `scale_pos_weight` (~16x) for XGBoost, equivalent to telling the model that each fraudulent row is 16x more costly to misclassify.
- **High cardinality text columns:** Zero-shot NLI classification (`cross-encoder/nli-MiniLM2-L6-H768`) was applied to title, company_profile, description, requirements, and benefits to produce a single interpretable legitimacy score per column. This avoids hundreds of sparse TF-IDF columns. A lightweight model was chosen over `facebook/bart-large-mnli` for speed on 12,000+ rows.
- **Clustering explored and rejected:** KMeans clustering on sentence embeddings for requirements and benefits showed no meaningful clusters, the elbow curve had no clear bend and the UMAP plot showed no visual separation. NLI scoring was adopted instead.
- **Logistic Regression convergence:** The model hit its iteration limit on the larger dataset. Features were scaled using StandardScaler and `solver='saga'` was used to improve convergence.

Evaluation & Results

Models were evaluated on the held-out 20% test set using three metrics: **Recall** (primary, minimise missed fraud), **F1-score** (balance of precision and recall), and **AUC-ROC** (overall discriminative ability across all thresholds).

Model	Recall (Fraud)	F1 (Fraud)	AUC-ROC
Random Forest	0.56	0.71	0.98
XGBoost	**0.80**	**0.79**	**0.98**
Logistic Regression	0.74	0.28	0.86

Table 1: Model performance on held-out test set. XGBoost (highlighted in green) is the recommended model.

Recommended model: XGBoost. It achieves the best balance across all three metrics, AUC of 0.98 (near-perfect discrimination), recall of 0.80 (catches 80% of fraudulent jobs), and F1 of 0.79. Random Forest matches on AUC but misses nearly half of all fraudulent jobs (recall 0.56). Logistic Regression has a reasonable recall (0.74) but a very low F1 of 0.28, indicating poor precision and excessive false alarms. XGBoost is the recommended model for deployment.

Top Predictive Features (XGBoost)

The XGBoost feature importance plot reveals the following top signals:

Feature	Importance	Insight
country_US	0.109	US postings carry strong signal
company_profile_legitimacy_score	0.083	Vague/missing profile is suspicious
required_education_High School or equivalent	0.065	Low barrier targets vulnerable seekers
country_MY	0.058	Malaysia postings are notable
required_education_Bachelor's Degree	0.048	Legitimate jobs specify qualifications

Table 2: Top 5 XGBoost feature importances with interpretation.

Notably, `country_US` is the top feature (0.109), suggesting that US-based job postings have a distinct pattern that the model finds highly discriminative, likely because the majority of both fraudulent and legitimate postings originate from the US, making geographic distribution an informative signal. `company_profile_legitimacy_score` (0.083) and `required_education_High School or equivalent` (0.065) follow closely, confirming that lack of professional credibility and low education barriers remain strong fraud indicators. `has_company_logo` ranks 7th (0.036), still a meaningful signal but less dominant on the full dataset compared to the balanced subset.

3. Dataset Preparation

Cleaning Steps

- Dropped rows missing `location`, job postings without any location are considered ambiguous in legitimacy.
- Dropped rows where *all* of employment_type, required_experience, required_education, industry, and function were null simultaneously (`how='all'`), retaining rows with partial information.
- Stripped HTML from company_profile, description, requirements, and benefits using BeautifulSoup; replaced newline characters with spaces.
- Parsed location into country/region/city; dropped region and city due to high cardinality (79 and 190 unique values respectively).
- Dropped department (~70% null). Filled required_education nulls with 'Unspecified'. Filled salary_range nulls with 'Unknown'.

Feature Engineering

- **NLP legitimacy scores:** Applied `cross-encoder/nli-MiniLM2-L6-H768` zero-shot NLI to generate a confidence score (0-1) that each text column belongs to the 'legitimate' label. Candidate labels were carefully worded to be descriptive, e.g. 'standard professional employee benefits' vs 'vague exaggerated benefits designed to lure victims'.
- **One-hot encoding:** Applied to employment_type, required_experience, required_education, and country (all low cardinality).
- **Label encoding:** Applied to industry (72 unique) and function (32 unique) to avoid excessive dimensionality. Tree-based models handle label-encoded categoricals well without implying false ordinal relationships.
- **Salary feature extraction:** Parsed salary_min, salary_max, and salary_mid from salary_range strings using regex. Added `has_salary` binary flag for missing salary entries.
- **Binary encoding:** telecommuting, has_company_logo, has_questions, and fraudulent converted from t/f strings to 0/1 integers.

Final Feature Set

The final dataset comprised **14,939 rows** and **122 features**: 5 NLP legitimacy score columns (company_profile_legitimacy_score, title_legitimacy_score, description_believability, requirements_reasonable_score, benefits_legitimacy_score), 4 binary columns (has_company_logo, telecommuting, has_questions, has_salary), 3 salary features (salary_min, salary_max, salary_mid), 2 label-encoded categorical columns (industry, function), 5 one-hot encoded employment type columns, 7 required experience columns, 13 required education columns, and 83 country columns, totalling 122 features.

Dataset: Recruitment Scam, Kaggle (amruthjithrajvr) | Models: scikit-learn, XGBoost | NLP: HuggingFace Transformers (cross-encoder/nli-MiniLM2-L6-H768)